



Qniverse hands-on webinar

Variational Quantum Eigensolver (VQE)

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Motivation

Why VQE?

- The problem of finding the **ground state energy** is ubiquitous in physics, chemistry, and related areas.
- Classical computers struggle:
 - Exact diagonalization works only for **small systems**.
 - Complexity grows **exponentially** with system size.
- Quantum computers offer a natural way to represent quantum states.
- Current state of quantum computing : **Noisy Intermediate Scale Quantum (NISQ) era**
- Need hybrid algorithms

VQE being a hybrid quantum–classical algorithm can leverage power of both, quantum and classical hardware.

This makes VQE one of the most promising algorithms for the **NISQ era**, enabling practical quantum advantage in areas like physics, chemistry, materials, and optimization.

VQE : An overview

To quote from “**Introduction to Quantum Mechanics**” (Chapter 7) by **David J. Griffiths**:

Suppose you want to calculate the ground-state energy E_g for a system described by the Hamiltonian H , but you are unable to solve the (time-independent) Schrödinger equation. Pick *any normalized function ψ whatsoever*.

Theorem:

$$E_g \leq \langle \psi | H | \psi \rangle \equiv \langle H \rangle. \quad [7.1]$$

The **variational principle** (as Equation 7.1 is called) is extremely powerful, and embarrassingly easy to use. What a chemist does, to find the ground-state energy of some complicated molecule, is write down a trial wave function with a large number of adjustable parameters, calculate $\langle H \rangle$, and tweak the parameters to get the lowest possible value. Even if ψ has no relation to the true wave function, one often gets miraculously accurate values for E_g . Naturally, if you have some way of guessing a *realistic* ψ , so much the better. The only *trouble* with the method is that you never know for sure how close you are to the target—all you can be *certain* of is that you’ve got an *upper bound*. Moreover, the technique applies only to the ground state (see.

VQE : An overview

Let H be the Hamiltonian of a quantum system and the ground state energy be E_0 (the **lowest eigenvalue** of H). For any normalized trial state $|\Psi\rangle$:

$$\langle \Psi | H | \Psi \rangle \geq E_0$$

This is the **variational principle**: the expectation value of H , with any trial state is always greater than or equal to the true ground state energy.

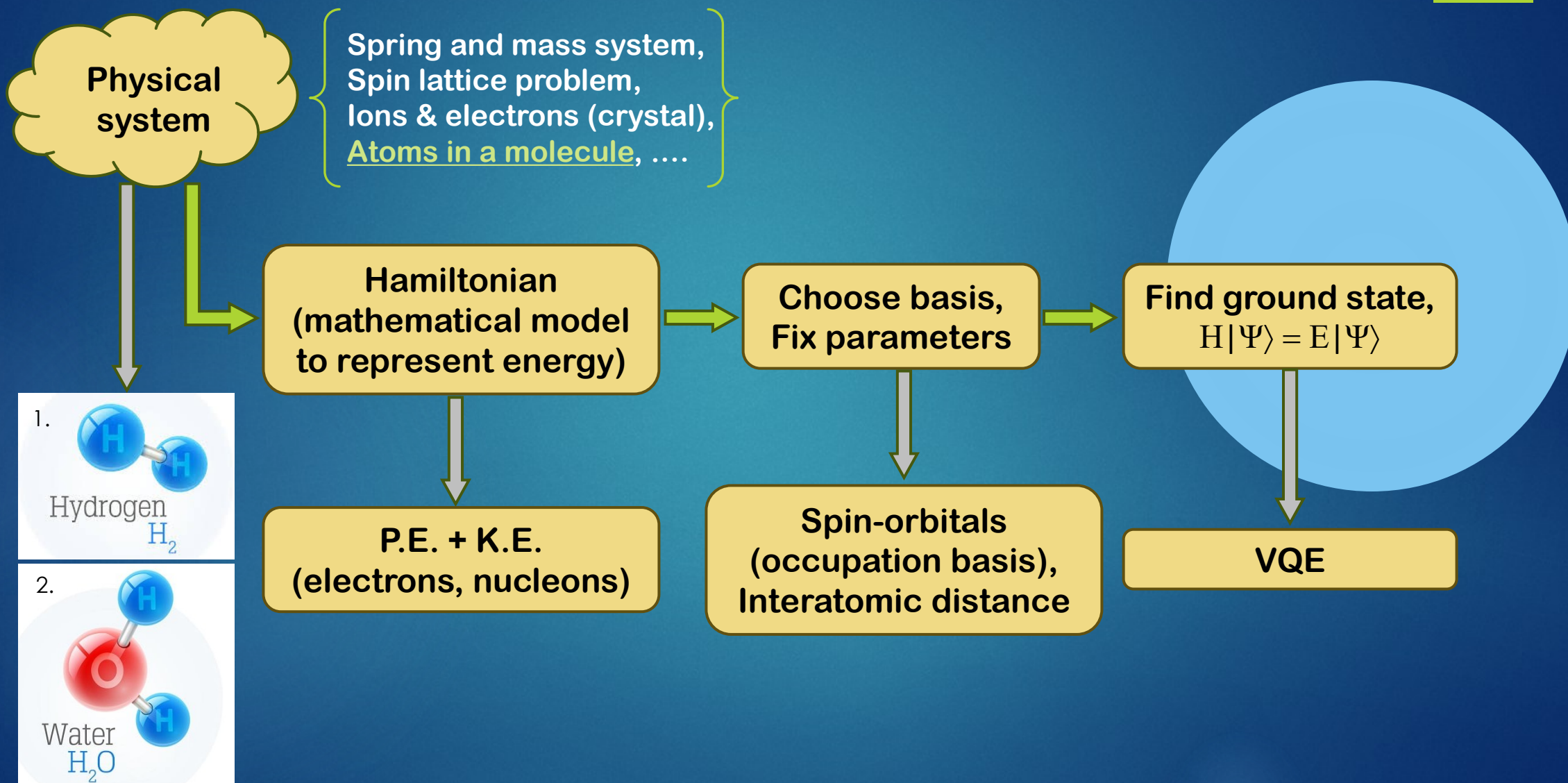
For a given trial wavefunction, say $|\Psi\rangle = |\Psi(\theta)\rangle$, the VQE algorithm optimizes the upper bound of the ground state energy as follows:

$$E_{VQE} = \min_{\theta} \frac{\langle \Psi(\theta) | H | \Psi(\theta) \rangle}{\langle \Psi(\theta) | \Psi(\theta) \rangle} \geq E_0$$

where θ is the parameter which is to be adjusted.

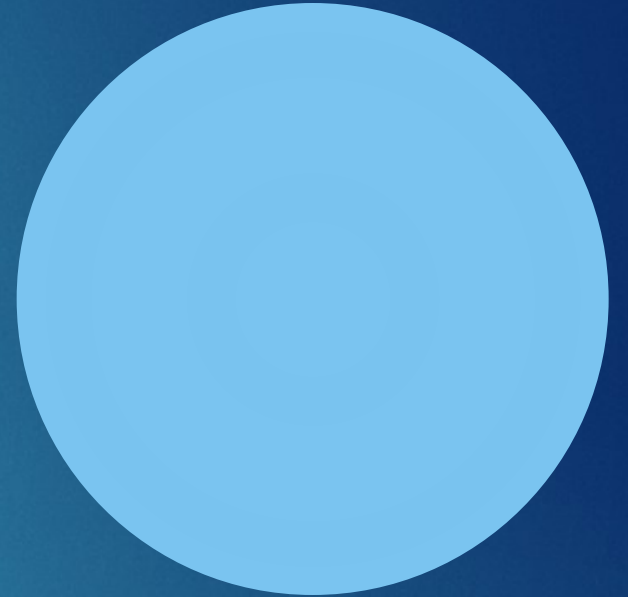
While the VQE was initially developed to find ground states of Hamiltonians, we can use it to find the minimum of any given objective function that we can express in a quantum circuit.

In general...



Formulation of the problem

Hamiltonian
(mathematical model
to represent energy)



Formulation of the problem

Hamiltonian
(mathematical model
to represent energy)

The electronic Hamiltonian, within Born-Oppenheimer approximation, is given as :

$$\hat{H} = - \sum_i^N \frac{1}{2} \nabla_i^2 + \sum_{A < B}^M \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|} - \sum_{i,A}^{N,M} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i < j}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

Here,

i, j (A, B) : denote electrons (nuclei),

$\mathbf{r}_i$ ($\mathbf{R}_A$) : denote position of an electron (nucleus),

Z_A, Z_B : charge numbers,

N (M) : number of electrons (nuclei).

Terms (left to right) : (i) kinetic energy of the electrons, (ii) the potential energy of nuclear–nuclear, (iii) nuclear–electron, and (iv) electron–electron interactions.

Formulation of the problem

Usually, the wave function for such systems is expressed in the form of Slater determinant* :

$$\psi(x_1, x_2, \dots, x_n) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_i(x_1) & \chi_j(x_1) & \cdots & \chi_k(x_1) \\ \chi_i(x_2) & \chi_j(x_2) & \cdots & \chi_k(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_i(x_n) & \chi_j(x_n) & \cdots & \chi_k(x_n) \end{vmatrix}$$

where, $\frac{1}{\sqrt{N!}}$ is the normalization parameter, and $\chi_m(x_l)$ represent the molecular orbital wavefunction of the m^{th} orbital of the l^{th} electron. The variable x_l contains both, spatial (coordinates) as well as spin information.

The Slater determinant naturally satisfies two necessary properties for electrons (fermions):

1. Wave function is anti-symmetric under particle exchange
Interchange of any two electrons is same as interchanging any two rows, which results in a sign-change in determinant, that is, $\psi(x_1, x_2, \dots) = -\psi(x_2, x_1, \dots)$
2. Pauli's exclusion principle
No two electrons can occupy the same orbital. This will end up with two χ 's (columns) becoming identical, resulting value of determinant to be 0.

The wavefunction above is often expressed as $|\chi_1\chi_2\chi_3 \dots\rangle$ for brevity.

*This is approximate (but good), as exact electron-electron correlations are not included.

Formulation of the problem

For H₂ molecule :

In Hydrogen, each atom consists of 1 *1s* orbital (hence 1 basis function). Therefore in H₂, we have a total of 2 spin orbitals (alongwith 2 spin contributions), resulting in 4 spin orbitals. If we wish to express the wavefunction in terms of Slater determinant then one has to construct determinants by picking up any 2 spin orbitals out of 4 at a time, e. g.

$$\psi(x_1, x_2) = \frac{1}{\sqrt{2!}} \begin{vmatrix} \chi_i(x_1) & \chi_j(x_1) \\ \chi_i(x_2) & \chi_j(x_2) \end{vmatrix}$$

and then consider the linear combination of all such 4 Slater determinants.

Formulation of the problem

Second quantized form : A more convenient approach

- Change Slater determinant notation to Occupation number (Fock) basis
The wavefunction is then expressed as :

$$|\psi\rangle = \sum c_i |n_1 n_2 n_3 \dots n_M\rangle, \quad n_i \in \{0, 1\}$$

where M denotes the number of spin-orbitals (N occupied orbitals + virtual orbitals).
The value of n_i then tells us if electron is in that state or not.

- The anti-symmetry inherent to fermionic systems is captured through operator properties :

$$a_p^\dagger |n_1, \dots, n_p, \dots, n_M\rangle = \delta_{n_p,0} (-1)^{\sum_{q<p} n_q} |n_1, \dots, (n_p + 1), \dots, n_M\rangle \quad (\text{creation op.})$$

$$a_p |n_1, \dots, n_p, \dots, n_M\rangle = \delta_{n_p,1} (-1)^{\sum_{q<p} n_q} |n_1, \dots, (n_p - 1), \dots, n_M\rangle \quad (\text{annihilation op.})$$

which also obey the following anticommutation relations:

$$\{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0 \quad \text{and} \quad \{a_p, a_q^\dagger\} = \delta_{pq}$$

The $(-1)^{\sum_{q<p} n_q}$ term handles the asymmetry ; Kronecker delta handles Pauli's exclusion principle.

Formulation of the problem

Second quantized form : A more convenient approach

If in case of Hydrogen molecule one represents the 4 orbitals as :

$$\phi_1 = (1s_A, \uparrow), \phi_2 = (1s_A, \downarrow), \phi_3 = (1s_B, \uparrow), \phi_4 = (1s_B, \downarrow)$$

then depending on whether they are unoccupied or occupied one can assign values 0 or 1 and write down a suitable representation, e.g.

$$|1100\rangle, |1010\rangle, |0110\rangle, \dots$$

Formulation of the problem

Mapping the Second quantized Hamiltonian to Qubits

- The Hamiltonian then looks like :

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$$

- Here, h_{pq} and h_{pqrs} represent the one- and two-electron integrals, respectively.
- Next important step : map the problem (state, operators) into Qubits using a suitable transformation (we use Jordan-Wigner transformation in the following discussion).

Make use of Pauli matrices (operators) $\{I, \sigma^x, \sigma^y, \sigma^z\} = \{I, X, Y, Z\}$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Formulation of the problem

Mapping the Second quantized Hamiltonian to Qubits

- Under Jordan-Wigner mapping :

$$a_n^\dagger \rightarrow \frac{1}{2} \left[\prod_{j=0}^{n-1} -Z_j \right] (X_n - iY_n) \quad \text{and} \quad a_n \rightarrow \frac{1}{2} \left[\prod_{j=0}^{n-1} -Z_j \right] (X_n + iY_n)$$

- With the help of the above and operators mentioned earlier, the Hamiltonian (for Hydrogen molecule) translates to :

$$\begin{aligned} \hat{H} = & b_1 \hat{I} + b_2 \hat{Z}_1 + b_3 \hat{Z}_2 + b_4 \hat{Z}_3 + b_5 \hat{Z}_4 \\ & + b_6 \hat{Z}_1 \hat{Z}_2 + b_7 \hat{Z}_1 \hat{Z}_3 + b_8 \hat{Z}_1 \hat{Z}_4 \\ & + b_9 \hat{Z}_2 \hat{Z}_3 + b_{10} \hat{Z}_2 \hat{Z}_4 + b_{11} \hat{Z}_3 \hat{Z}_4 \\ & + b_{12} (X_1 Y_2 Y_3 X_4 + Y_1 X_2 X_3 Y_4 - X_1 X_2 Y_3 Y_4 - Y_1 Y_2 X_3 X_4) \end{aligned}$$

*coefficients $b_1 \dots b_{12}$ depend on the internuclear distance.

VQE : Implementation

As stated earlier, before using VQE we need to parameterize our quantum circuit and choose a suitable trial wavefunction (Ansatz) as well. Some of the popular choices are **EfficientSU2**, **HardwareEfficient**, **UCCSD (Unitary Coupled Cluster with Singles and Doubles)**. The choice depends upon the problem in picture. In the following we will be using **UCCSD**.

On UCCSD :

Coupled Cluster theory is a powerful quantum chemistry tool which can be used for calculations beyond mean-field. It takes into account strong electron correlations and is systematically improvable by including higher order corrections (here, excitations). It is sufficient to use it with Singles and Doubles excitations in the following.

Reference State : Start from Hartree–Fock state $|\Psi_0\rangle$

Excitation Operators:

Singles : promote one electron $\rightarrow$ another orbital

Doubles : promote two electrons $\rightarrow$ two other orbitals.

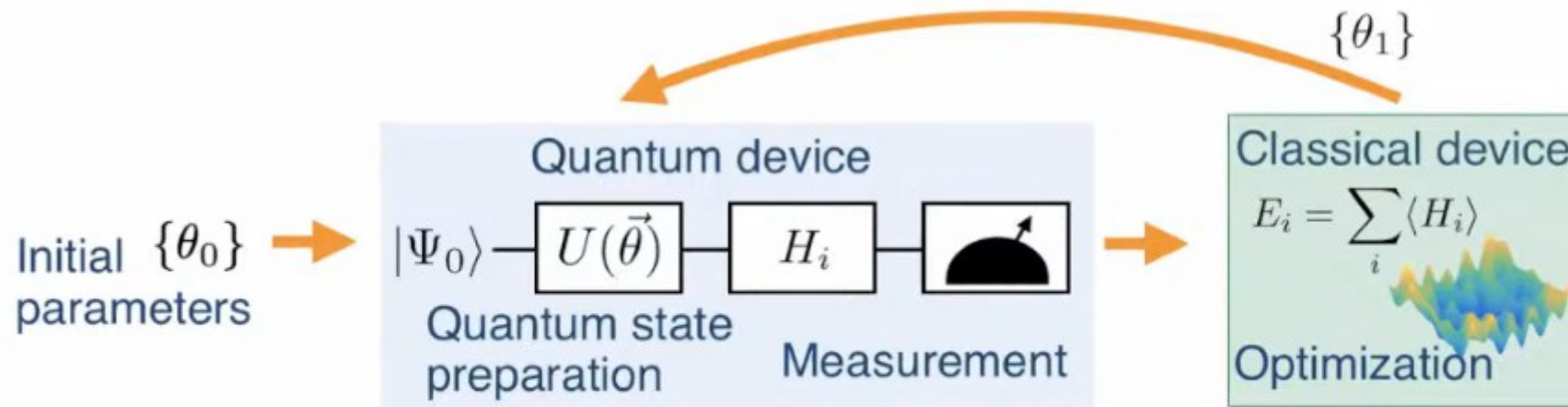
VQE : Pipeline

The Variational Quantum Eigensolver

1. **Quantum** state preparation (use parameteric ansatz to describe the wavefunction)
2. **Classical** Optimization (variational principle)

$$\Psi(\vec{\theta}) = U(\vec{\theta})|\Psi_0\rangle$$

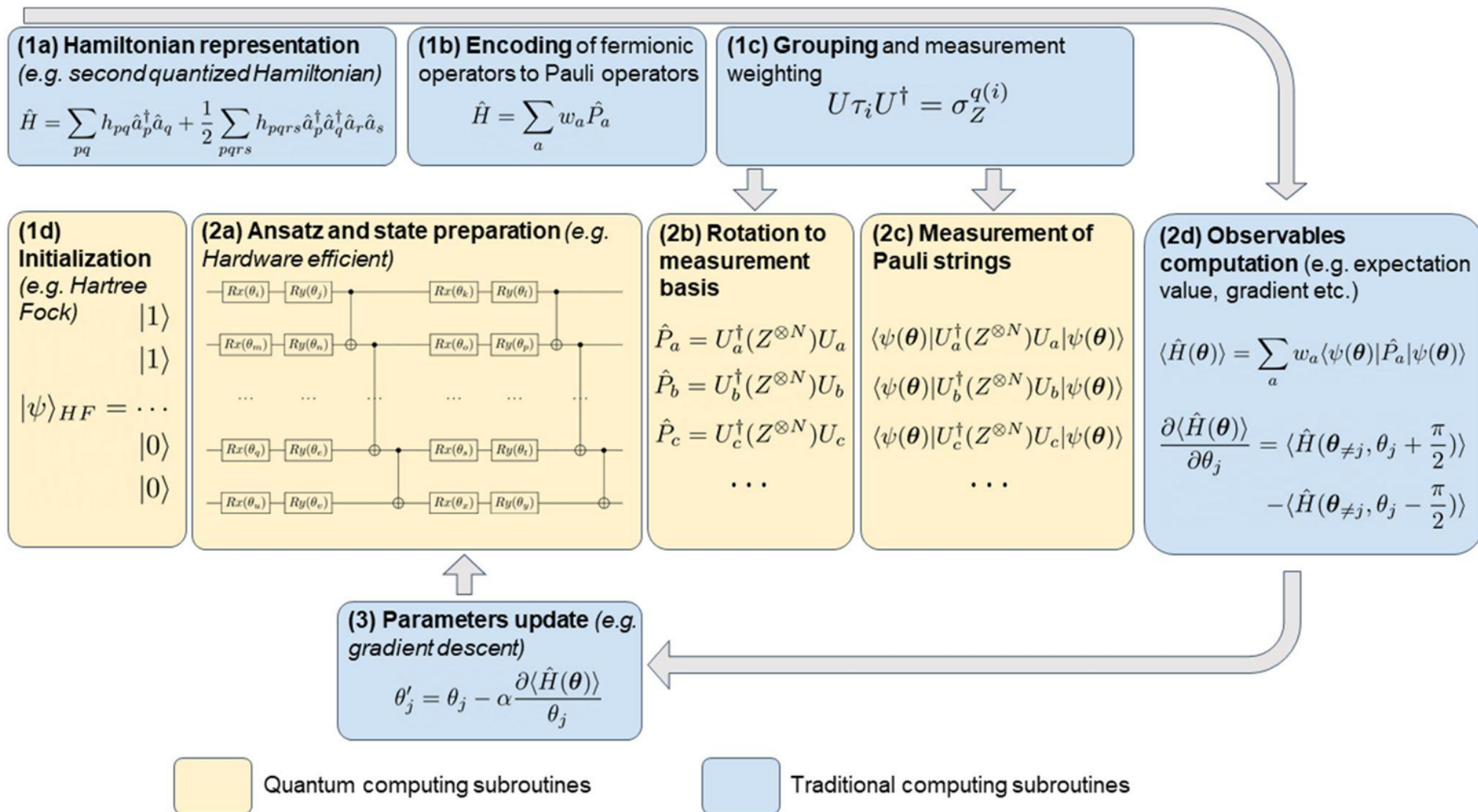
$$E = \min_{\vec{\theta}} \langle \Psi(\vec{\theta}) | \hat{H} | \Psi(\vec{\theta}) \rangle \geq E_{\text{exact}}$$



VQE : $|\Psi(\theta)\rangle = U(\theta)|\Phi_0\rangle$; $|\Phi_0\rangle$ is known

$E(\theta) = \langle \Phi_0 | U^\dagger(\theta) H U(\theta) | \Phi_0 \rangle$ U : gates / $|\Phi_0\rangle$: Qubits

VQE : Pipeline



VQE : Pipeline

1.A) Hamiltonian construction and representation

Define the system. Basis selection and Hamiltonian representation.

1.B) Encoding of operators

Encode Hamiltonian into a Quantum Computer compatible format (fermionic operators to qubits)

1.C) Measurements Strategy and Grouping

To get accurate expectation values, we must run the quantum circuit many times, each ending with a measurement. Our goal is to **reduce the number of repetitions** needed.

To save resources, operators are often **grouped together** so they can be measured at the same time. This usually requires adding a rotation to the circuit, so that all operators in the group become diagonal in the chosen measurement basis.

1.D) Ansatz and state preparation

Preparation of the trial wave function; decide on an ansatz.

VQE : Pipeline

D 1.1) The VQE Loop

1.1 (a) Start with qubits in the initial state. Apply the ansatz circuit with some starting parameters.

1.1 (b) Rotate the state into the right measurement basis (generally Z basis by default). Perform measurements to get expectation values of the Hamiltonian terms.

1.1 (c) Combine the measurement results (e.g., weighted sum) to calculate the energy.

1.1 (d) Use a classical optimizer to update the ansatz parameters based on the measured energy. Repeat the loop until the energy converges to the ground state value.

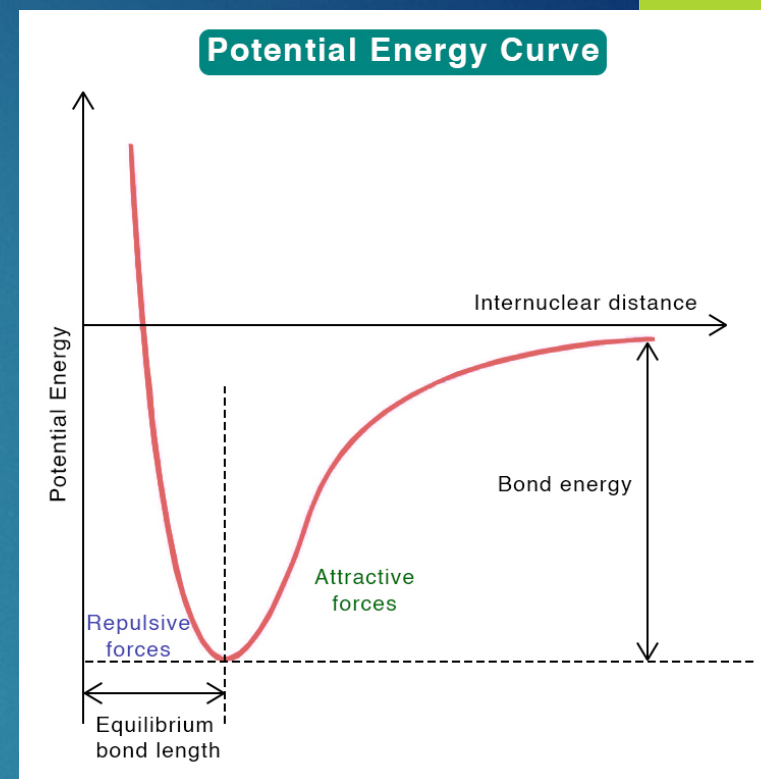
The H₂ molecule problem

When the individual Hydrogen atoms are substantially far apart they are attracted by van der Waals like forces.

If they are too close the repulsive forces (due to same charge interaction) dominate.

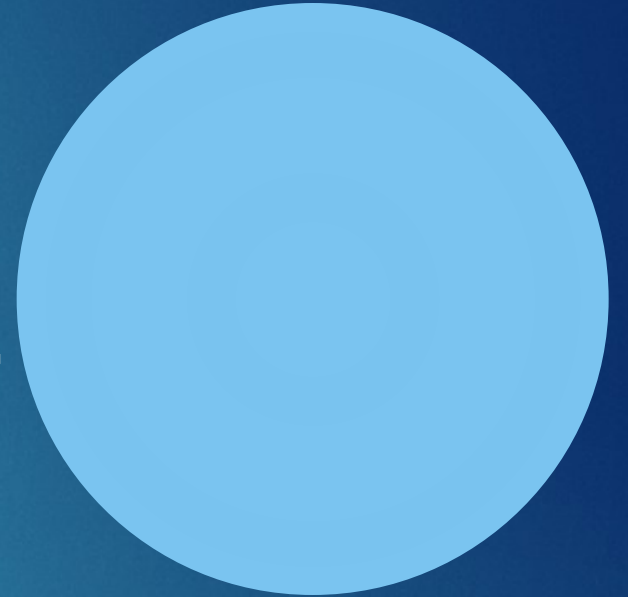
The problem is to find the equilibrium position (interatomic distance) at which molecule formation can take place (if that is possible).

This situation is depicted in the graphical form here ==>



In the following simulation we will vary the interatomic distance systematically and calculate the energy at each step. A negative value will naturally indicate a bound state and the interatomic distance at which the energy is the minimum will give us the interatomic separation (stable configuration) for Hydrogen molecule.

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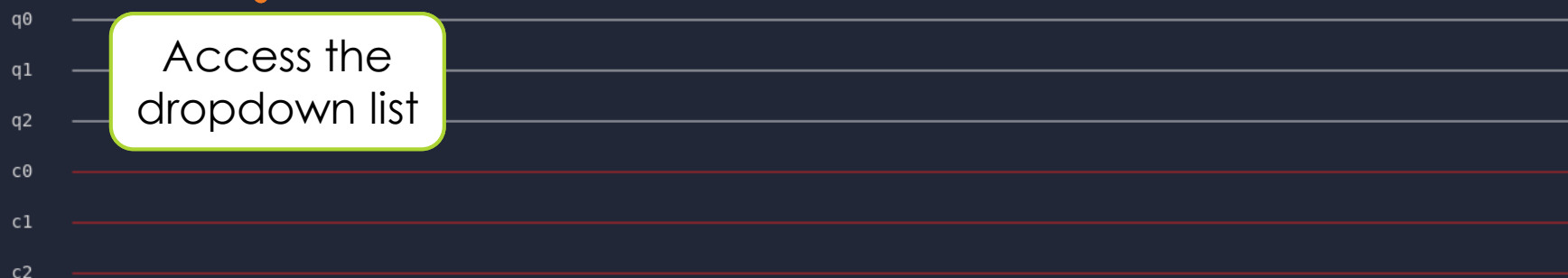
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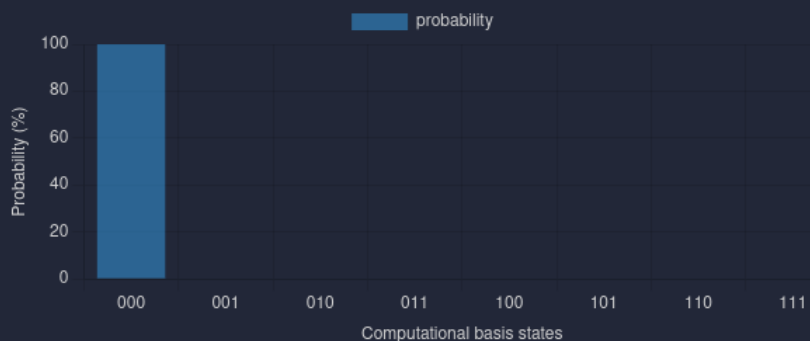


Access the dropdown list

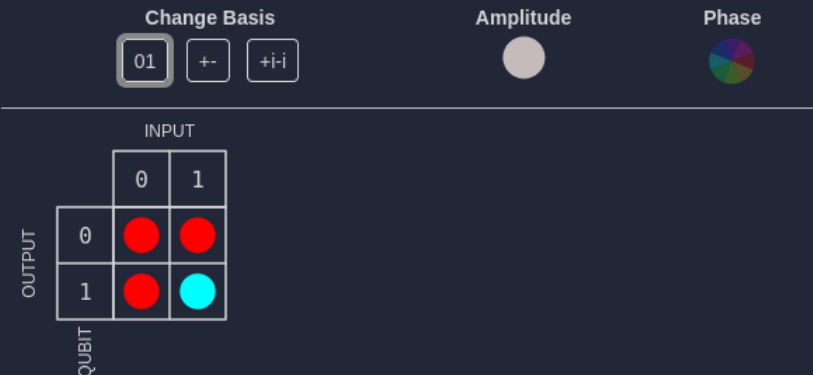
OpenQASM2.0

```
1 OPENQASM 2.0;
2 include "qelib1.inc";
3
4 qreg q[3];
5 creg c[3];
```

Probability Distribution



Hadamard gate



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Circuit ID	Status
6503	✓
6502	✓
6021	✗
6020	✗
6019	✓

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Hadamard gate



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probability

Change Basis

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+/-

+i-i

Amplitude



Phase



Circuit ID

Status

7961



Superdense coding is a quantum communication protocol that allows sending two classical bits of information by transmitting only one qubit. This is achieved by leveraging a shared entangled pair of qubits between the sender and receiver, doubling the classical communication capacity per qubit sent.

Enter input bit

Load Circuit

The Quantum Fourier Transform (QFT) is the quantum equivalent of the classical Discrete Fourier Transform. It can be implemented efficiently on a quantum computer and is a fundamental subroutine used in many powerful quantum algorithms, including Shor's factoring algorithm and Quantum Phase Estimation.

2 Qubits

Load Circuit

Quantum Phase Estimation (QPE) estimates the phase ϕ of a unitary operator U for its eigenvector $|\psi\rangle$, where $U|\psi\rangle = e^{2\pi i \phi} |\psi\rangle$. It provides a precise estimate of ϕ . This is a key component used within many quantum algorithms, most notably Shor's algorithm for factoring.

Load Circuit

Shor's Algorithm - Factorization

Shor's algorithm is a quantum algorithm for finding the prime factors of a composite integer N . It offers a dramatic, exponential speedup over the best-known classical factoring algorithms, posing a significant threat to widely used classical encryption methods like RSA.

Enter Number

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A 1D Quantum walk is the quantum analogue of a classical random walk. Unlike the classical diffusive spread ($\sqrt{t}$), the quantum walk spreads quadratically faster (t), useful for quantum search algorithms and other applications.

2-Qubits

Load Circuit

Variational Quantum Eigensolver (VQE)

The Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm. It uses a quantum computer to prepare and measure quantum states, while a classical optimizer minimizes the energy expectation value of a Hamiltonian, commonly applied to finding molecular ground state energies.

Molecule :

BeH₂

Load Circuit

Harrow-Hassidim-Lloyd (HHL) Algorithm

The Harrow-Hassidim-Lloyd (HHL) algorithm is a quantum algorithm designed to solve linear systems of equations of the form $Ax = b$. HHL offers an exponential speedup in the matrix dimension N compared to classical methods, under the conditions that the matrix A is sparse, well-conditioned, and Hermitian.

Load Circuit

Veto Algorithm

Quantum Veto is a protocol that uses quantum entanglement to anonymously block a decision. If even one person blocks it, without revealing who did it. The process ensures participant anonymity.

(This algorithm is inspired from the work of Satish Kumar and Anirban Pathak. <https://doi.org/10.1007/s11128-022-03650-2>)

Load Circuit

OSVM Algorithm

BeH₂
H₂O
H₂
LiH
N₂
CH₄
HF
NH₂
NH₄
RbH

used for classification and minimize the hyperplane mass separation.

Load Circuit

QKmean Algorithm

Quantum K-Means algorithm is an innovative advancement in quantum computing and machine learning that harnesses quantum mechanics

QPC for the socialist millionaire problem

Quantum Private Comparison(QPC) algorithm enables secure value comparison using entangled Bell pairs, quantum rotations, and a shared key,

QKNN

Quantum k-Nearest Neighbors adapts classical kNN to quantum computing leveraging quantum superposition and the swap test for parallel state

Scroll down to find
'Variational Quantum Eigensolver'

Select H₂ from
the dropdown list

Superdense Coding

Superdense coding is a quantum communication protocol that allows sending two classical bits of information by transmitting only one qubit. This is achieved by leveraging a shared entangled pair of qubits between the sender and receiver, doubling the classical communication capacity per qubit sent.

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Load Circuit

Select VQE Preference

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Load

Enter the co-ordinates of both the atoms in (x,y,z) format (Unit : Å) and Load the algorithm

quadratically faster (t), useful for quantum search algorithms and other applications.

2-Qubits

Load Circuit

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Molecule :

H₂

Load Circuit

QSVM Algorithm

QSVMs are quantum machine learning algorithms used for classification and regression that leverage quantum computers to optimize the hyperplane selection in high-dimensional spaces for maximal class separation.

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H Y Z T S P T[†] S[†]
RX RY RZ

U ⊕ ⊗ I X CZ CP CY
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VQE algorithm is loaded for H₂

VQE-Algo Manage Undo Redo

Clear Debug

q0

Circuit generates only with QASM code. It is not available for editable qiskit.

c0

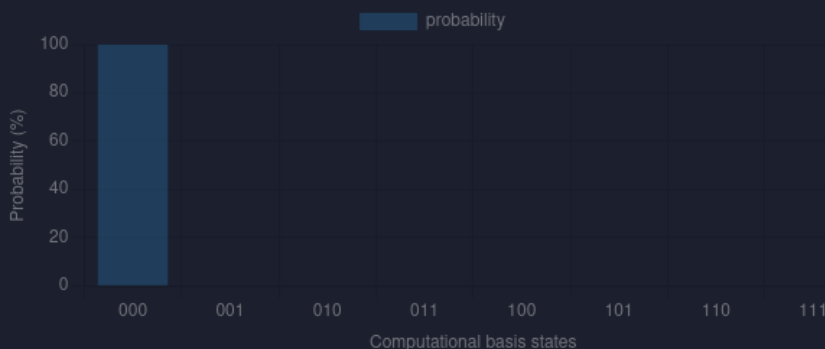
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c2

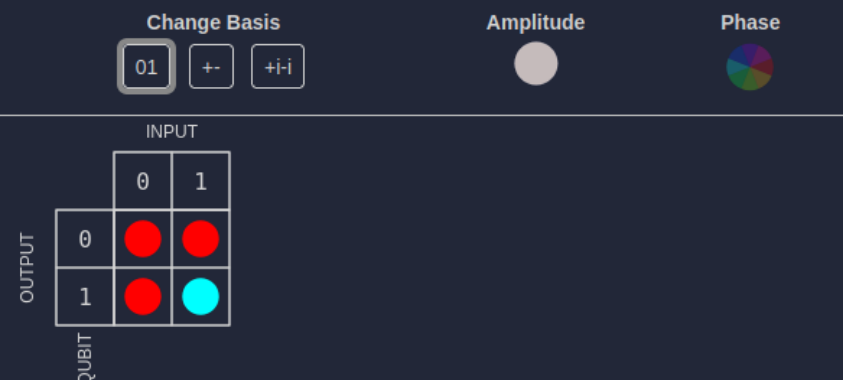
Qiskit (Editable)

```
1 from qniverse.algorithms import MolecularEnergy
2 from qiskit_nature.second_q.formats.molecule_info import MoleculeInfo
3 from qiskit_nature.second_q.drivers import PySCFDriver
4 from qiskit_nature.second_q.transformers import FreezeCoreTransformer
5 molecule = MoleculeInfo(
6     symbols=["H","H"],
7     coords=([0,0,0], [1.689,0,0]),
8     multiplicity=1,
9     charge=0
10 )
11 driver = PySCFDriver.from_molecule(molecule)
12 properties = driver.run()
13 problem = FreezeCoreTransformer(freeze_core=True).transform(properties)
14 vqe = MolecularEnergy()
```

Probability Distribution



Hadamard gate



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Circuit ID	Status
6503	✓
6502	✓
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6020	✗
6019	✓

Code overview

```
from qniverse.algorithms import MolecularEnergy
```

→ for setting up the 'problem', exact solution, VQE solution

```
from qiskit_nature.second_q.formats.molecule_info import MoleculeInfo
```

→ molecule's atomic structure (symbols, coordinates, charge, multiplicity)

```
from qiskit_nature.second_q.drivers import PySCFDriver
```

→ Utilize PySCF (Python-based Simulations of Chemistry Framework) to analyze the molecular structure. Pass the defined molecule to the PySCFDriver, which will compute the molecular integrals and return essential electronic structure properties.

```
from qiskit_nature.second_q.transformers import FreezeCoreTransformer
```

→ This reduces the number of orbitals by freezing core electrons and optionally removing selected orbitals. Returns a reduced electronic structure problem optimized for simulation.

```
molecule = MoleculeInfo(...)
```

→ setting up the molecule

Code overview

```
driver = PySCFDriver.from_molecule(molecule)
```

→ compute integrals and properties of the defined molecule

```
properties = driver.run()
```

→ runs the PySCF calculation to obtain molecular electronic properties

```
problem = FreezeCoreTransformer(freeze_core=True).transform(properties)
```

→ Transforms the molecular problem by freezing core orbitals

```
vqe = MolecularEnergy()
```

→ Initializes the molecular energy calculation object for VQE

```
vqe_result, final_result = vqe.compute_molecule_energy_from_problem(problem)
```

→ VQE computation of the ground-state energy of the molecular problem

Other important parameters / values

Molecule Info : distance unit is Angstrom (10^{-10}m)

Basis : STO-3G (Slater-Type Orbital – 3 Gaussian function) basis

It's a minimal basis set we can use to approximate the atomic orbitals (others : 6G, etc.). Here each STO is expressed approximately by a linear combination of 3 Gaussian functions.

Parity Mapper

To convert the molecular Hamiltonian into a qubit representation.

Ansatz

For molecular simulations, UCCSD is generally more chemically accurate and better suited, so we choose it as ansatz in our simulation.

Optimizer and Estimator

We use SLSQP (Sequential Least Squares Programming) optimizer, gradient based method, suited for VQE like problems (others : COBYLA, SPSA, or L-BFGS-B)

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S[†]

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QFT

IQFT

<>

VQE-Algo

Manage

↶ Undo

↷ Redo

↶ Clear

Debug

q0

!

Circuit generates only with QASM code. It is not available for
editable qiskit.

c0

c1

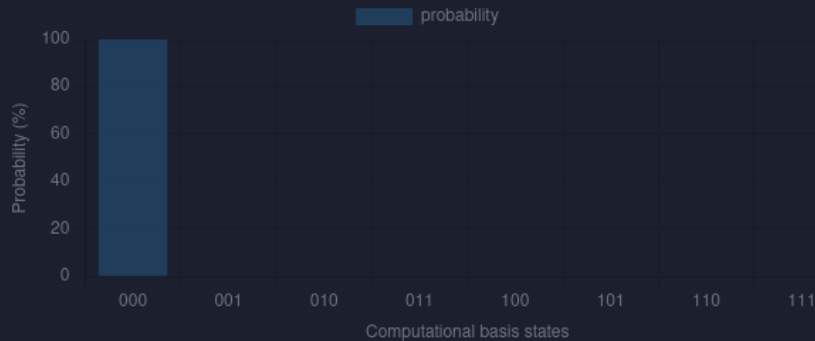
c2

Qiskit (Editable)

```

1 from qniverse.algorithms import MolecularEnergy
2 from qiskit_nature.second_q.formats.molecule_info import MoleculeInfo
3 from qiskit_nature.second_q.drivers import PySCFDriver
4 from qiskit_nature.second_q.transformers import FreezeCoreTransformer
5 molecule = MoleculeInfo(
6     symbols=["H", "H"],
7     coords=([0,0,0], [0.7,0,0]),
8     multiplicity=1,
9     charge=0
10 )
11 driver = PySCFDriver.from_molecule(molecule)
12 properties = driver.run()
13 problem = FreezeCoreTransformer(freeze_core=True).transform(properties)
14 vae = MolecularEnergy()
```

Probability Distribution



Hadamard gate

Change Basis

01

+-

+i-i

Amplitude

●

Phase

●

INPUT

	0	1
0	●	●
1	●	●

OUTPUT

	0	1
0	●	●
1	●	●

QUBIT

Latest Jobs

Circuit ID	Status
7961	⚙️
7960	✓
7959	✓
7943	✓
7942	✓

✓ JOB ID: 7961

Hello Manpreet!

We are on a mission to help Researchers & Students
in exploring quantum circuits and algorithms

Circuit ID:

7941

Circuit Name:

VQE-Algo 7941

Resource:

QACC_Cluster-CPU

Backend:

Status

COMPLETED

Submitted using Editable Qiskit:

Yes



Created: Sep 4, 2025 11:49 AM



In queue

NA



Running

usage: 00:00:21.000000



Completed

Refresh

Load Circuit

Job
status

Histogram



Histogram Not Available for Editable Qiskit

Output

Error

Output
window

Final ground state energy using exact solver (Classical) ==> -0.9726277801401663

Iteration 1: Parameters: [0.0, 0.0, 0.0], Energy: -1.1705678046226664

Iteration 2: Parameters: [1.0, 0.0, 0.0], Energy: -1.036646102305443

Iteration 3: Parameters: [0.0, 1.0, 0.0], Energy: -1.036646102305443

Iteration 4: Parameters: [0.0, 0.0, 1.0], Energy: -0.6750769722359541

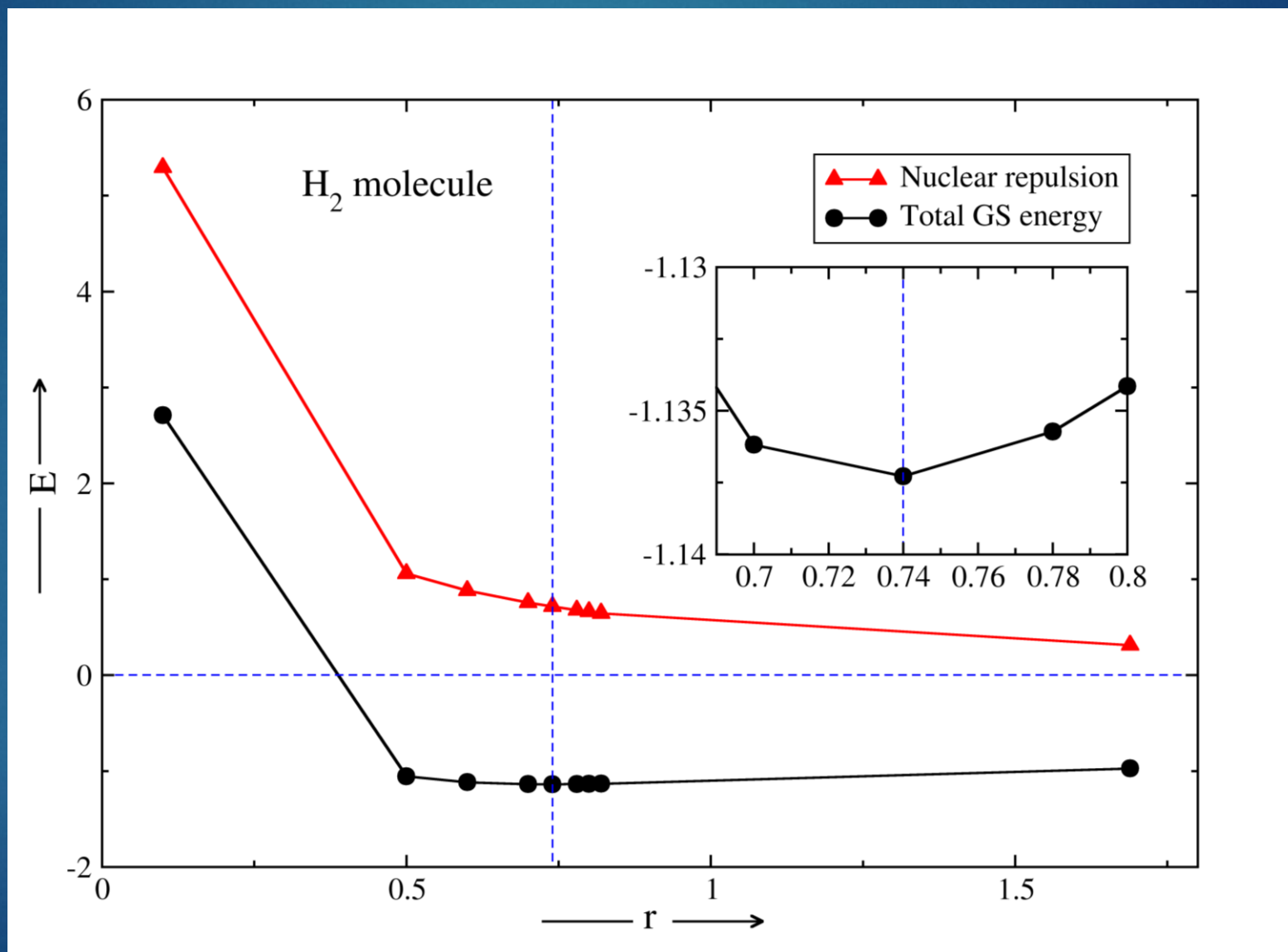
Iteration 5: Parameters: [-0.2524662997767925, -0.2524662997767925, -0.9340886119389474], Energy: -0.12623314988839626, -0.12623314988839626, -0.4670443059694737], Energy: -0.4118893924713254, -0.4118893924713254, -0.7616644379136527], Energy: 0.0505435454082406, -0.30300984518503316, -0.4670443059694737], Energy: 0.1569578045240019, 0.1724101875491714, -0.7509721847400271], Energy: -0.29234223375623697, -0.26726587693850323, -0.5895903877403597], Energy: -0.10647717748031166, -0.09732476797562667, -0.3470484435016517], Energy: -0.046555611331001445, -0.037838354821018536, -0.5052915288023953], Energy: 0.018667851418894085, 0.05780228245679765, -0.5524483778066087], Energy: -0.09342663207915212, 0.0034458879634833364, -0.5075205895321049], Energy: 0.04286175787037759, 0.03642467179205125, -0.5512756234805637], Energy:

Circuit

QASM

Transpiled Code

Sample results



H₂O molecule

The screenshot displays the Qniverse web application interface. The top navigation bar includes the Qniverse logo, a Help menu, and user information (Balance: 100, Manpreet). The main content area shows a quantum circuit diagram with qubits q0, q1, and q2. The circuit includes a Hadamard gate on q0, followed by CNOT gates between q0 and q1, and q1 and q2. The circuit is labeled "Circuit ID: 7961" and "Circuit Name: VQE-Algo 7961". The resource is "QACC_Cluster-CPU" and the backend is "COMPLETED". The status is "Submitted using Editable Qiskit: Yes".

The right panel displays the "Hello Manpreet!" message and a "Refresh" button. Below this, a "Histogram" tab is selected, showing a message: "Histogram Not Available for Editable Qiskit". The "Output" tab is also visible, showing the final ground state energy using an exact solver (Classical) as -74.97742821123101 . The output also lists parameters for 15 iterations.

The bottom of the interface shows a "Terms & Conditions" link and a "Transpiled Code" button. The URL in the address bar is qniverse.in/QACC/detailedreport.php?jobid=7961#.

Useful references

1. Book : “Introduction to Quantum Mechanics” by David J. Griffiths
2. Qniverse documentation : <https://qniverse.in/docs/variational-quantum-eigensolver-vqe/>
3. arXiv paper : <https://arxiv.org/pdf/2305.06538>
4. Notes on VQE : The Variational Quantum Eigensolver in Chemistry by Stefania-R. Symeonidou
5. Other : IBM Qiskit documentation and Tutorials

Thank You!

